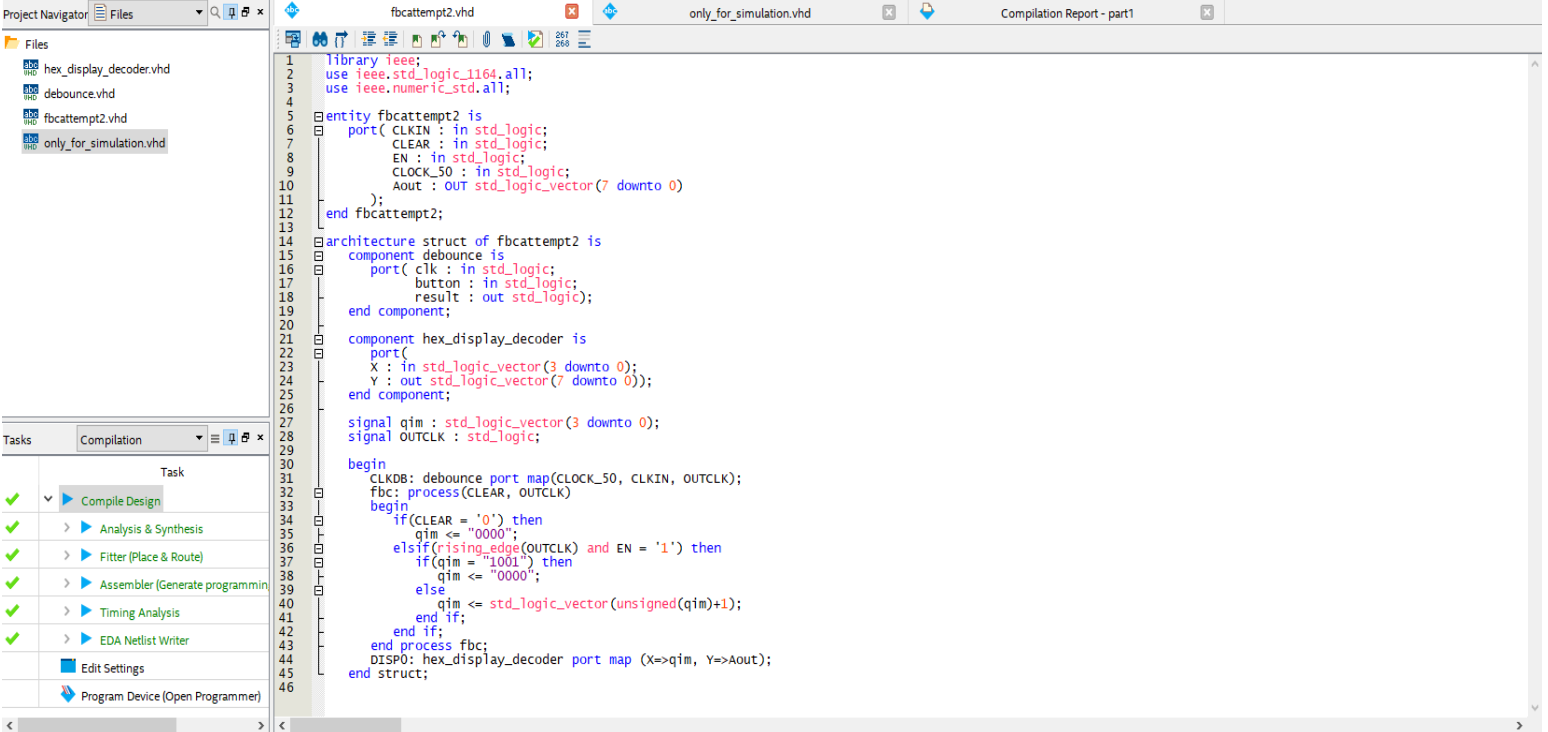


在模拟顶层实体的代码时，不要声明 debounce 相关组件，并且不要实例化和端口映射 debounce，否则模拟的时序图曲线将不正确（输出会处于 unknown 状态）。可以在模拟仿真完成后，再添加 debounce 部分的代码功能；

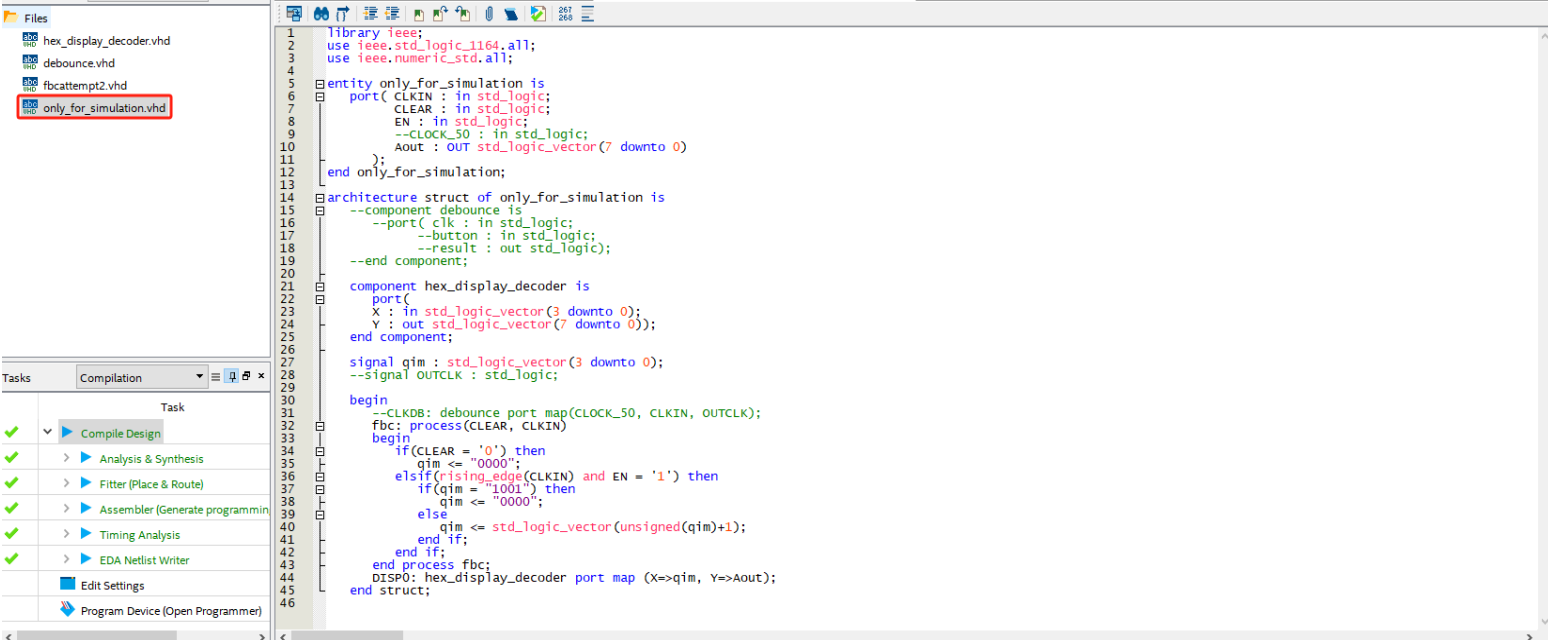
如果你已经在顶层实体中添加了 debounce 相关的代码，你可以采用注释的方式（或者你也可以再重新新建一个顶层实体仅用于仿真），并且需要修改部分已经实例化的代码；

原因是，仿真无法调用 FPGA 开发板的硬件资源，而 debounce 相关功能使用了系统的 50Mhz 内部时钟作为一个输入端口，而 debounce 的输出信号与输入有关

下两张图有两段代码，分别是用于烧录和用于仿真的，供理解。



```
1 library ieee;
2 use ieee.std_logic_1164.all;
3 use ieee.numeric_std.all;
4
5 entity fbcattempt2 is
6 port(
7     CLKIN : in std_logic;
8     CLEAR : in std_logic;
9     EN : in std_logic;
10    CLOCK_50 : in std_logic;
11    Aout : out std_logic_vector(7 downto 0)
12 );
13 end fbcattempt2;
14
15 architecture struct of fbcattempt2 is
16     component debounce is
17         port(
18             clk : in std_logic;
19             button : in std_logic;
20             result : out std_logic;
21         );
22     end component;
23
24     component hex_display_decoder is
25         port(
26             X : in std_logic_vector(3 downto 0);
27             Y : out std_logic_vector(7 downto 0);
28         );
29     end component;
30
31     signal qim : std_logic_vector(3 downto 0);
32     signal OUTCLK : std_logic;
33
34 begin
35     CLKDB: debounce port map(CLOCK_50, CLKIN, OUTCLK);
36     fbc: process(CLEAR, OUTCLK)
37     begin
38         if(CLEAR = '0') then
39             qim <= "0000";
40         elsif(rising_edge(OUTCLK) and EN = '1') then
41             if(qim = "1001") then
42                 qim <= "0000";
43             else
44                 qim <= std_logic_vector(unsigned(qim)+1);
45             end if;
46         end if;
47     end process fbc;
48     DISP0: hex_display_decoder port map (X=>qim, Y=>Aout);
49 end struct;
```



```
1 library ieee;
2 use ieee.std_logic_1164.all;
3 use ieee.numeric_std.all;
4
5 entity only_for_simulation is
6 port(
7     CLKIN : in std_logic;
8     CLEAR : in std_logic;
9     EN : in std_logic;
10    --CLOCK_50 : in std_logic;
11    Aout : out std_logic_vector(7 downto 0)
12 );
13 end only_for_simulation;
14
15 architecture struct of only_for_simulation is
16     --component debounce is
17     --port(
18     --    clk : in std_logic;
19     --    button : in std_logic;
20     --    result : out std_logic;
21     --);
22     --end component;
23
24     component hex_display_decoder is
25         port(
26             X : in std_logic_vector(3 downto 0);
27             Y : out std_logic_vector(7 downto 0);
28         );
29     end component;
30
31     signal qim : std_logic_vector(3 downto 0);
32     --signal OUTCLK : std_logic;
33
34 begin
35     --CLKDB: debounce port map(CLOCK_50, CLKIN, OUTCLK);
36     fbc: process(CLEAR, CLKIN)
37     begin
38         if(CLEAR = '0') then
39             qim <= "0000";
40         elsif(rising_edge(CLKIN) and EN = '1') then
41             if(qim = "1001") then
42                 qim <= "0000";
43             else
44                 qim <= std_logic_vector(unsigned(qim)+1);
45             end if;
46         end if;
47     end process fbc;
48     DISP0: hex_display_decoder port map (X=>qim, Y=>Aout);
49 end struct;
```